

Lab 3 – Architecture Selection & Justification

System: Podcast Guest Scheduling & Outline Builder

Architecture Selection: We chose Layered Architecture for the Podcast Guest Scheduling & Outline Builder system.

Architectural Choice: The system separates presentation, business/application, and data/integration responsibilities. The User Interface Component handles interaction with podcast guests and show hosts, while the Scheduling & Booking, Outline Management, and Run-of-Show Generator components implement the main business functions. Data and external calendar interactions are handled separately.

Reason 1 – Separation of Concerns: Layered architecture separates user interaction, business processing, and data/integration responsibilities. This makes scheduling, outline management, and run-of-show functionality easier to understand, maintain, and modify independently.

Reason 2 – Maintainability: The system can change one part without requiring major changes to unrelated components. For example, the Calendar Service integration can be modified without changing the guest interface or outline management functionality.

Security Advantage: Requests can pass through business components instead of allowing the user interface to directly access stored booking and guest information. This provides a controlled point for authentication and authorization.

Performance Benefit: Scheduling and outline operations can be handled within the application layer while data access is separated into the data/integration layer. This allows database and service interactions to be optimized independently and avoids unnecessary processing in the user interface.